

Evanns Morales-Cuadrado

Robotics PhD Candidate | U.S. Citizen | Fulbright Specialist

CONTACT

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SUMMARY

Robotics PhD candidate in Dr. Samuel Coogan's FACTS Lab at Georgia Tech and AI for Autonomy intern at Sandia National Laboratories. Specializes in safe autonomy, reinforcement learning, and real-time control. Proven track record of developing and deploying safe, computationally efficient planning and control methods for autonomous systems, with work published in leading peer-reviewed robotics and controls venues. Selected to the U.S. Department of State's Fulbright Specialist Roster after spearheading an international research collaboration with a Brazilian partner institution and securing research funding.

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, C#/.NET, Matlab/Simulink
- **Robotics & Control Frameworks:** ROS 2, NATS, PX4, ArduPilot, acados, CasADi
- **Machine Learning & Simulation:** JAX, PyTorch, MuJoCo, Reinforcement Learning
- **Development & DevOps Tools:** Docker, Continuous Integration, Git/GitHub
- **Hardware & Experimentation:** UAV assembly, repair, and deployment
- **Motion-Capture:** OptiTrack and Vicon
- **Languages:** Spanish (Native), English (Native), Portuguese (B1)

EDUCATION

Georgia Institute of Technology

2022-Present

Ph.D. in Robotics

Doctoral Minor in Optimization Methods

- GPA: 4.0/4.0
- Advisor: Dr. Samuel Coogan
- Thesis Proposal Approved (ABD)
- Goizueta Fellow

University of Texas at Arlington

2018-2022

Honors Bachelor of Science in Electrical Engineering

Minor in Mathematics

Minor in Physics

Certificate in Unmanned Vehicle Systems

- GPA: 3.9/4.0 Summa Cum Laude
- Full National Merit Scholarship Awarded
- Electrical Engineering Honors Scholar

WORK EXPERIENCE

Year-Round Graduate Research Intern | AI for Autonomy

May 2026 - Present

Sandia National Laboratories

Albuquerque, NM

- Applied reinforcement learning to autonomy problems in support of national security
- Designed and deployed robotics and automation infrastructure that enables multi-robot experiments
- Built C#/.NET networking frameworks and execution scripts that coordinate communication and task execution across multi-robot systems, now used team-wide

Graduate Research Assistant | FACTS Lab

August 2022 - Present

Georgia Institute of Technology

Atlanta, GA

- Advised by Dr. Samuel Coogan
- Developed a lightweight tracking control law that beat PX4's native tracker in aggressive quadrotor flight and matched NMPC's tracking accuracy at far lower computation time and CPU energy
- Developed safe planning and control methods that combine reachability analysis, formal methods, optimization, and learning to generate fast, provably safe trajectories under unknown disturbances
- Built the lab's real-time flight-test stack, enabling repeatable hardware validation

- Teaching Assistant** | Vertically Integrated Project in Robotics January 2024 - May 2026
Georgia Institute of Technology Atlanta, GA
- Mentored about 20 undergraduates per semester on hands-on robotics research, with multiple mentees going on to robotics PhD programs
 - Designed and delivered undergraduate lectures on ROS, Git/GitHub, Linux, and control theory, preparing students for research on robotics and autonomous control systems
- Research Assistant** | Autonomous Systems Lab November 2021 - May 2022
University of Texas at Arlington Arlington, TX
- Advised by Dr. Frank Lewis
 - Trained in one of the most-cited control research groups in electrical engineering, with Dr. Lewis' work ranked 12th in the USA and 23rd worldwide by citation impact
 - Contributed to reinforcement learning-based quadrotor control research
- Teaching Assistant** | Graduate-Level Intelligent Systems Course January 2022 - May 2022
University of Texas at Arlington Arlington, TX
- Selected as an undergraduate to TA a graduate-level course
 - Led exam review sessions and graded homework, exams, and projects
- Research Assistant** | Dynamical Networks and Control Lab October 2019 - May 2022
University of Texas at Arlington Arlington, TX
- Advised by Dr. Yan Wan
 - Completed UTA Honors College thesis on learning-based minimum-time and minimum-energy multi-vehicle path planning

RESEARCH INTERESTS

- Safe Autonomy
- Hardware Deployment
- Unmanned Ground and Aerial Vehicles
- Advanced Nonlinear Control
- Learning-Based Control
- Trajectory Generation and Planning

SELECTED PUBLICATIONS

- E. Morales-Cuadrado**, S. Coogan, Z. Kakish, and A. Parikh. Submitted to *2027 IEEE International Conference on Robotics and Automation (ICRA)*. (title redacted until the double-blind review process is complete)
- L. Pinto*, **E. Morales-Cuadrado***, S. Coogan, M. A. Wehrmeister, L. Caselatto, and C. E. Magrin. Submitted to *2027 IEEE International Conference on Robotics and Automation (ICRA)*. *Equal contribution. (title redacted until the double-blind review process is complete)
- M. Cao, A. Harapanahalli, **E. Morales-Cuadrado**, and S. Coogan, "Trajectory Tracking for Systems with Unknown Time-Varying Disturbances." *IEEE Open Journal of Control Systems (OJ-CSYS)*, 2026.
- E. Morales-Cuadrado**, "Computationally Efficient and Safe Control for Aerial Robotic Systems under Threat and Disturbance." Ph.D. Thesis Proposal, Georgia Institute of Technology, 2026. [pdf available](#). [slides available](#).
- E. Morales-Cuadrado**, L. Chung, S. Kousik, and S. Coogan, "RTD-RAX: Fast, Safe Trajectory Planning for Systems under Unknown Disturbances." Joint Submission to 2026 Modeling, Estimation and Control Conference (MECC) and ASME Letters in Dynamic Systems and Control (ALDSC). (accepted to both) [preprint available](#).
- E. Morales-Cuadrado**, L. Baird, Y. Wardi, and S. Coogan, "Lightweight Tracking Control for Computationally Constrained Aerial Systems with the Newton-Raphson Method." *IEEE Transactions on Control Systems Technology (TCST)*, 2026. [preprint available](#).
- L. Baird, **E. Morales-Cuadrado**, and S. Coogan, "Runtime Assurance for Uncertain Systems from Interval Signal Temporal Logic." Submitted to IFAC Nonlinear Analysis: Hybrid Systems (NAHS), 2025. (under revision) [preprint available](#).

E. Morales-Cuadrado, C. Llanes, Y. Wardi, and S. Coogan, “Newton-Raphson Flow for Aggressive Quadrotor Tracking Control.” *2024 American Control Conference (ACC)*. [preprint available](#).

OPEN-SOURCE CONTRIBUTIONS

- **PX4 Autopilot:** Contributed documentation improvements to the official PX4 Autopilot manual.
- **Vicon4PX4:** Developed and open-sourced a ROS 2 C++ Vicon–PX4 interface enabling external vision fusion for indoor quadrotor flight experiments; adopted and showcased by Georgia Tech’s Indoor Flight Lab as part of its official codebase. Also created a parallel package for OptiTrack systems.

SERVICE & LEADERSHIP

- Co-Founder and President of Puerto Rican Student Association** August 2024 - Present
Georgia Institute of Technology Atlanta, GA
- Addressed the need for an organization to help Puerto Rican students feel at home and stay connected to our culture
 - Formed a three-member co-founder board and identified a faculty advisor
 - Raised funds for the organization and recruited members from campus
 - Hosted professional, social, and cultural events to meet the needs of the growing Puerto Rican student population at Georgia Tech
- Undergraduate Research Mentor** August 2021 - May 2022
University of Texas at Arlington Arlington, TX
- As the lead undergraduate student in Dr. Frank Lewis’ Autonomous Systems Lab, mentored junior undergrads in controls research and guided them on a research project in ground vehicle control
 - Guided students in Robot Operating System, Python, Linux, and Control Systems
- Eta Kappa Nu Public Relations Chair & Circuits I Tutor** August 2020 - May 2022
University of Texas at Arlington Arlington, TX
- Executive board member of UT Arlington’s IEEE-HKN chapter
 - In charge of social media advertising for the honor society’s events and outreach initiatives for the electrical engineering student body
 - Volunteered to provide free tutoring as a service to electrical engineering students in Circuits I

HONORS AND AWARDS

- Fulbright Specialist** August 2026
Peer-review-selected to the U.S. Department of State’s Fulbright Specialist Roster for a three-year term
- Top-3 Finalist at a Deep Learning Research Symposium** November 2023
Professor-sponsored award for novel research in deep learning and robotics
- Goizueta Fellowship at Georgia Tech** August 2022
Highly selective fellowship for graduate students of Latin-American descent
- Summa Cum Laude Honors at University of Texas at Arlington** May 2022
Exceptional academic distinction in top GPA tier for graduation honors
- Honors College Degree Requirements at University of Texas at Arlington** December 2021
Completed research projects in disparate fields throughout my undergraduate career with various professors and defended a senior capstone research project before the Honors College committee
- Electrical Engineering Honors Scholar at University of Texas at Arlington** December 2021
Inaugural member of the electrical engineering honors cohort
- IEEE-HKN International Electrical Engineering Honor Society Membership** August 2021
Selective undergraduate society recognizing outstanding academic achievement and community service
- Research Experience for Undergraduates Fellowship Recipient** August 2020
Fellowship to sponsor promising young undergraduate researchers
- Chance Vought Engineering and Science Endowment Scholarship** August 2020
Highly selective award given annually to three recipients across the entire College of Engineering
- Toray Composite Materials America Inc. Sons and Daughters Scholarship** May 2019
Academic scholarship awarded to children of employees of Toray America

James E. Casey Scholarship Recipient Academic scholarship awarded to children of employees of UPS	January 2019
James R. Hoffa Memorial Scholarship Recipient Academic scholarship awarded to children of Teamsters Union Members	May 2018
Grapevine, TX Rotary Club Scholarship Recipient Local rotary club scholarship for academic and civically-minded scholars	May 2018
National Hispanic Scholar Recognition of top national performers of Latin descent on the SAT	March 2018
National Merit Scholar Recognition of top national performers on the SAT	March 2018